CSE 260: Digital Logic Design

Lab Report - 5

Experiment Name: Design and Implementation of 4 bit parallel Binary Adder

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Section: 22

Name of the Experiment: Design and Implementation of 4-bit Parallel Binary Adder.

Objective:

- To implement half-adder circuit.
- To implement full-adder circuit using two half adder circuits.
- To implement a four bit parallel adder.

Required Components and equipments:

- AT-700 portable analog/ digital laboratory
- Breadboard
- Wires
- AND gate
- OR gate
- X-OR gate
- 4 bit parallel adder

Experimental Setup

Part 1 : Half Adder Circuit

To implement the half adder, we need to take one XOR and one AND gate. ~~Firstly, we connect vcc to the pin~~ For each IC, we connect vcc to the pin 14 and ground to the pin 7. First we take two inputs from data switches to the XOR gate. The inputs will be inserted in pin 1 and pin 2. The output we get will work as sum of the two bits in pin 3. Secondly, the two inputs from the perivious switches will be again inserted on a AND IC. The pin 1 and 2 will be used as inputs and the output we get from pin 3 in the AND IC will be used to show carry. Thus, from XOR IC the output we get is $S = X \oplus Y$ which will later inserted in the LED display and eventually the output of the AND IC will show the carry in the LED display.

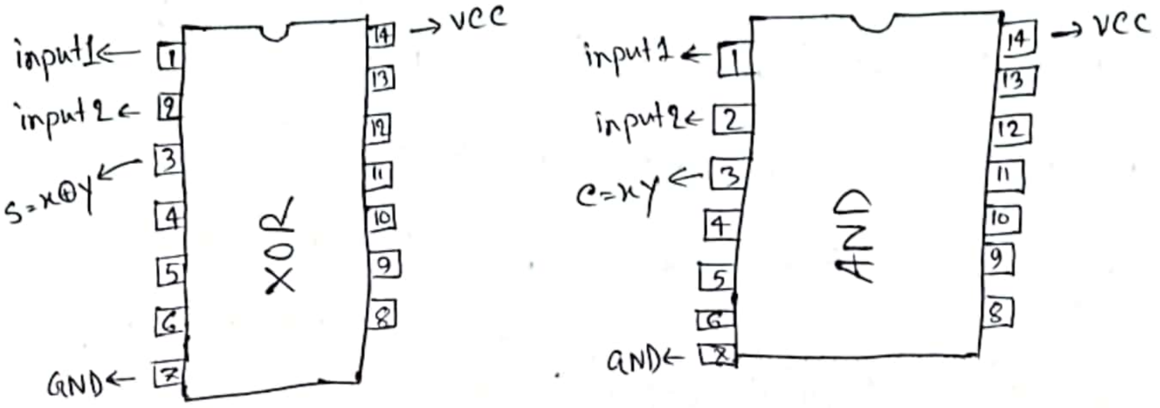


Fig: IC diagram

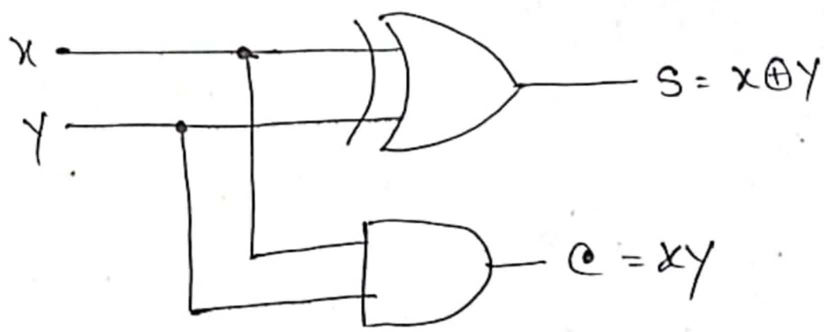


Fig: Half Adder circuit

Part 2: Full Adder Circuit

A full adder is a combinational circuit that forms the arithmetic sum of three input bits. To implement the full adder, we need to use another gate, OR gate. The existing half adder circuit will be used to make full adder. Firstly, the XOR gate output will be inserted into another ~~ga~~ XOR gate. Again, another input considered as ~~a~~ Z will be inserted as 2nd input to the X-OR gate. Thus, $(x \oplus y) \oplus z$ will be the output. So, we take the output of $X \oplus Y$ into pin 4 through a wire. Another input z ~~from~~ inserted in pin 5. The output we get from pin 6 will be used as sum.

Now, we will insert the output of pin 3 to another AND gate. The output will $z.(x \oplus y)$. So, pin 3 will be inserted in pin 4 and another input z will be inserted in pin 5. The output from pin 6 will be inserted in the OR gate. The output we will get $XY + z(X \oplus Y)$, which will be used to show carry.

Now, in the OR gate, in pin 1 and pin 2 we will insert the inputs we get from pin 6 of AND gate and pin 3 also from AND gate. The output we get from OR gate pin 3 will be used to show ~~et~~ carry.

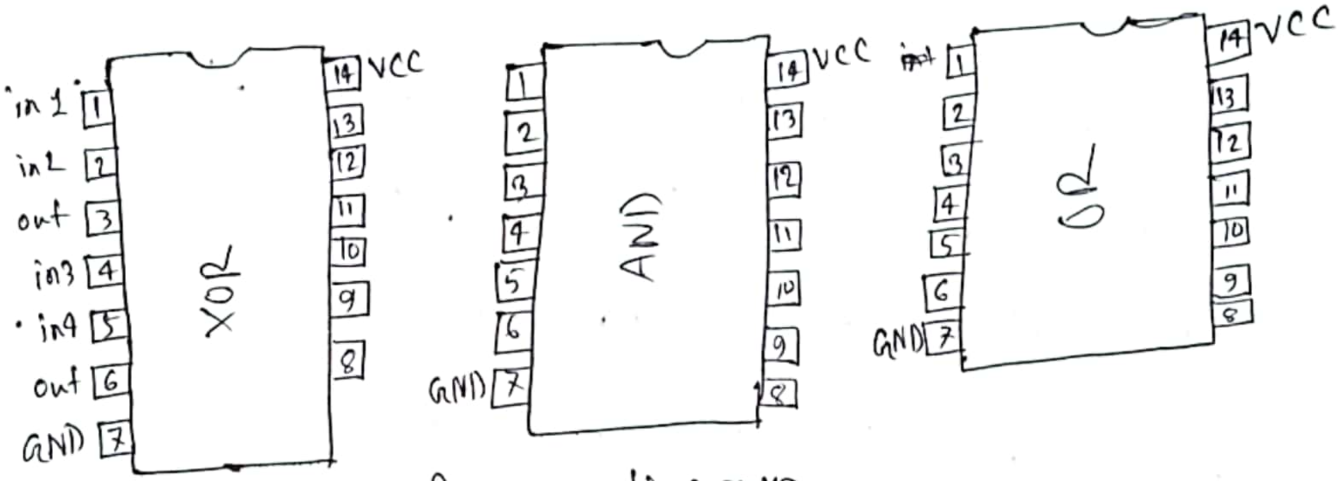


fig: IC diagram

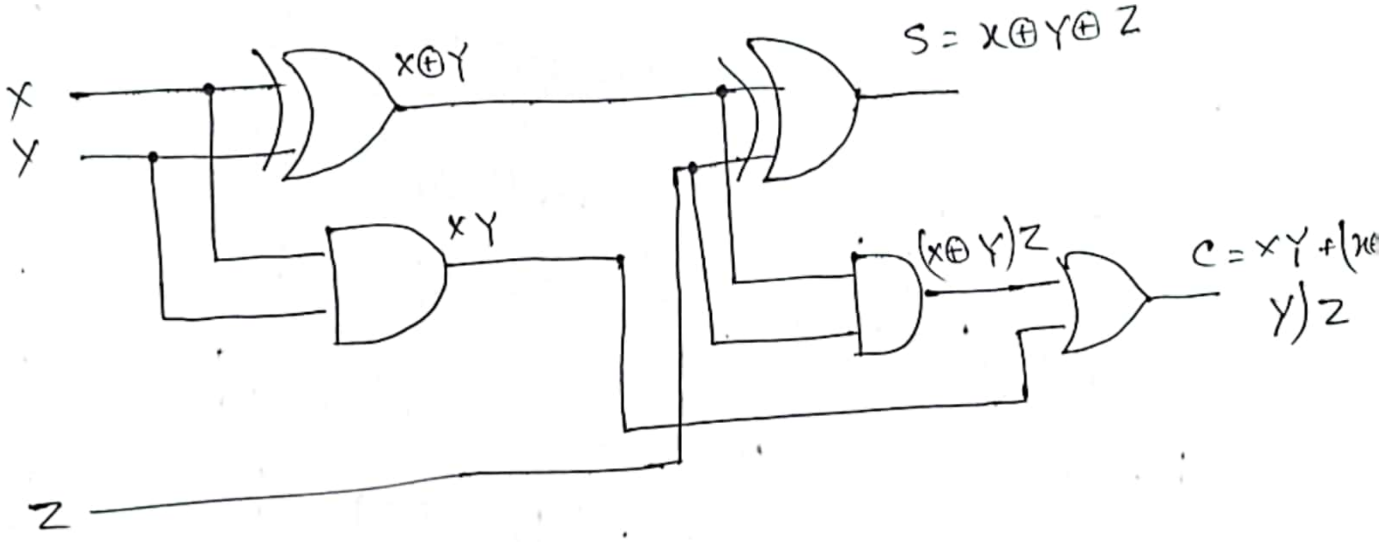


Fig: Full Adder circuit

~~Task-3~~

Part-3 : A four bit parallel adder

We will use a IC to implement four bit parallel adder, IC (7483). The IC is ~~a~~ different from the IC's we ~~are~~ used previously. To power up the IC, we need to connect the pin 5 to Vcc and pin 12 to GND. Our first carry is 0 so C_0 is connected to pin 13. A_4, A_3, A_2, A_1 ~~and A_0~~ will be inserted in pin 1, 3, 8, 10 ~~respectively~~ respectively. Again B_4, B_3, B_2, B_1 will be inserted to pin 16, 4, 7, 11 respectively. The LED display should be organised in a way that shows the carry in left most side. So, $C_4, \Sigma_4, \Sigma_3, \Sigma_2, \Sigma_1$ will be the outputs that are going to be inserted serially. We will take the outputs including $C_4, \Sigma_4, \Sigma_3, \Sigma_2, \Sigma_1$ from the pin of 14, 15, 2, 6, 9 respectively.

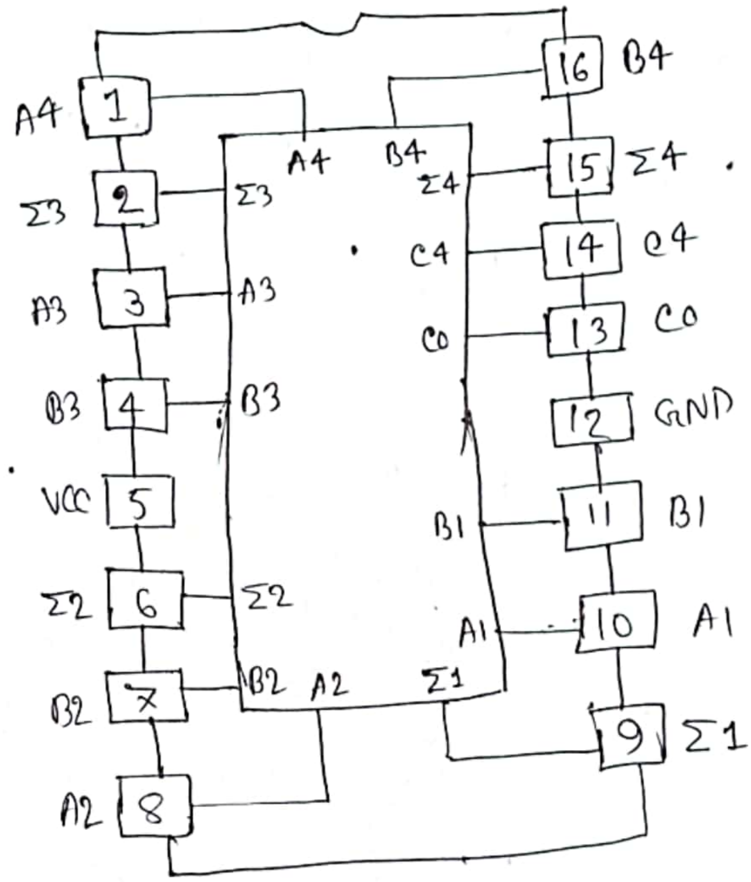
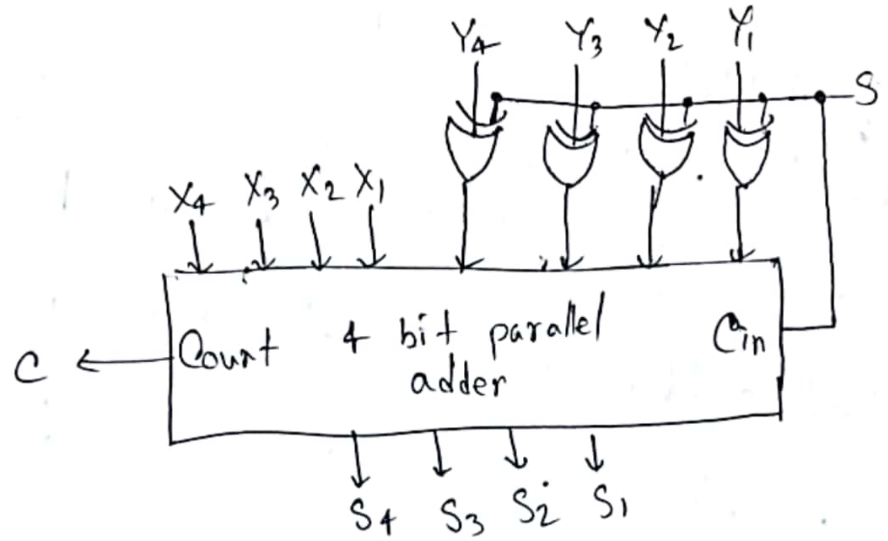


fig: IC diagram (IC - 7483)

Part 4 : A 4 bit parallel adder cum subtractor:

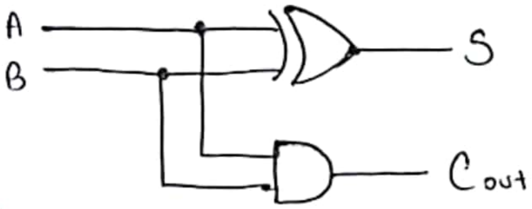
Now, we have to transform the 4 bit parallel adder to a 4 bit parallel adder substractor. A 4 bit parallel adder can only perform addition while 4 bit parallel adder - substractor can perform both addition or substraction depending on the initial carry bit. To convert the 4 bit parallel adder to a 4 bit parallel adder substractor we will use X-OR gate with Cin and all the ~~B~~ inputs. Like we will connect Y_1 and Cin using X-OR gate. We will do the same thing for all the ~~inp~~ Y inputs. While converting it to a 4 bit parallel adder substractor we will keep Cin constant for all. Now to perform addition we will keep our Cin 0 and to do substraction we will keep our Cin 1.



Results

Truth table for half adder:

A	B	Sum	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



$\text{Sum} = \bar{A}B + A\bar{B}$

$= A \oplus B$

$\text{Cout} = AB$

Truth table for full adder:

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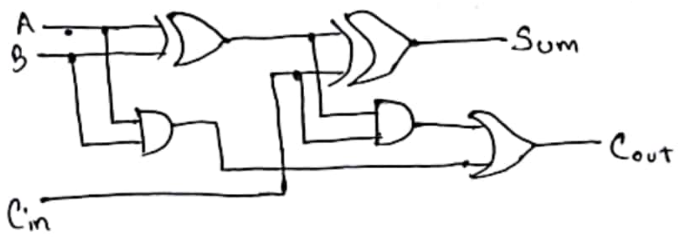
A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Using SOP:

$\text{Sum} = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$

$= \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}\bar{C} + BC)$

$= \bar{A}(B \oplus C_{in}) + A(\overline{B \oplus C_{in}})$

$= A \oplus (B \oplus C_{in})$

~~In 4 bit parallel adder substra· substractor circuit we ws~~

Using SOP :

$$C_{out} = ABC_{in} + ABC_{in}' + AB'C_{in} + A'BC_{in}$$

$$= C_{in}(AB' + A'B) + AB(C_{in} + C_{in}')$$

$$= C_{in}(A \oplus B) + AB$$

$$\begin{array}{r} 1100 \\ +\ 0011 \\ \hline 1111 \end{array} \qquad \begin{array}{r} 1110 \\ -\ 0011 \\ \hline 1011 \end{array}$$

X_4	X_3	X_2	X_1	Y_4	Y_3	Y_2	Y_1	C_{in}	S_4	S_3	S_2	S_1	C_{out}
1	1	0	0	0	0	1	1	0	1	1	1	1	0
1	1	1	0	0	0	1	1	1	1	0	1	1	0

Therefore, the theoretical values of addition and substraction match with the circuit implementation.

Discussion

To design and implement the 4 bit parallel binary adder, we used X-OR gates to connect C_{in} with Y_4, Y_3, Y_2, Y_1 to invert the binary inputs of Y_4, Y_3, Y_2, Y_1. Then the rest of the calculation is done by the full adder.

Throughout the experiment, we faced problem with the jumper wires. Many of our jumper wires were defected. Thus we got wrong output many times and also it took longer time to complete the experiment as we had to check every wires if it was working or not.